

Pranav Singh

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Research Interests: Neural Algorithmic Reasoning, Geometric Deep Learning, ML for Algorithm Design, Reinforcement Learning, GNN Expressivity & Size Generalization, Equivariant Learning

EDUCATION

- **University of California San Diego** Sep 2023 - Jun 2027
Bachelor of Science in Data Science La Jolla, CA
 - **Mathematics & Computer Science:** Real Analysis, Optimization, Abstract Algebra, Mathematical Reasoning, Random Matrix Theory, Linear Algebra, Vector Calculus, Theoretical Foundations of Data Science, Intro to Algorithms
 - **Machine Learning:** Geometry of Data, Reinforcement Learning, Representation Learning, Probabilistic Modeling & Machine Learning, Large Language Models, Natural Language Processing, Recommender Systems

RESEARCH EXPERIENCE

- **University of California San Diego, Wang Lab** Dec 2025 - Present
Research Assistant, Neural Algorithmic Reasoning for Hierarchical Clustering La Jolla, CA
 - Designed a neurally aligned architecture for learnable hierarchical clustering, preserving Lance-Williams updates while optimizing over standard and mixed linkage metrics
 - Built a permutation and scale-invariant point encoder across modality types, experimenting with architectures balancing global and local features
 - Proved a lower bound of $5/4$ on the approximation ratio of complete linkage for L_1 diameter-sum via an explicit four-point instance, and showed the learned linkage recovers the optimal tree on it
 - Fine-tuned with GRPO and reward shaping, ablating adapters across four insertion points; added Gumbeldore self-improvement for DP-free teacher generation via stochastic beam search with logit steering
 - Showed OOD transfer on real chip layouts, where a ranker trained on uniform distributions beat complete linkage on **58%** of samples drawn from 62 OpenROAD clock-sink designs
- **Yale University, Graph and Geometric Learning Lab** Dec 2025 - May 2026
Research Assistant, Hierarchical Tree Construction & Spectral Clustering Remote
 - Built a bottom-up hierarchical tree construction pipeline clustering document embeddings into a bounded-branching tree, using spectral clustering over the pairwise similarity graph rather than flat k -means
 - Implemented Nyström-approximated spectral clustering, clustering in the eigenspace of a landmark similarity graph for $O(nm)$ memory instead of $O(n^2)$, cutting memory **50%** and scaling construction from 50K to **450K+** documents
 - Extended clustering to Euclidean and Lorentz embedding spaces, cosine spectral clustering against geodesic k -means, and evaluated hierarchical retrieval across **15** tree configurations spanning three geometries to test how embedding geometry shapes tree structure
- **University of California San Diego** May 2025 - Present
Ledell Research Scholar, Heterogeneous Graph Learning for Pharmacological Knowledge Graphs La Jolla, CA
 - Adapted GNN-LLM architectures to use heterogeneous GNNs with type-aware embeddings, reaching **0.44** Hit@1 on pharmacological graph-QA (**+0.32** over baseline) by encoding relation types as an inductive bias
 - Developed the PharmQA benchmark: **4,000+** subgraphs sampled from an **87K**-node biomedical knowledge graph via relationship-aware random walks over **12** entity types for multi-hop GNN-LLM evaluation
- **National Nanotechnology Coordination Office** May 2024 - Aug 2025
Research Intern Remote
 - Developed an ontology-aware negative sampling scheme for link prediction on a **131,388**-node federal nanomaterial knowledge graph, constraining negatives to scientifically plausible experimental combinations
 - Built semantic mapping across EPA, NIOSH, and CPSC databases using ontology alignment and schema matching, with a query system over the unified graph
- **University of California San Diego, Chiba Lab** Sep 2024 - Present
Research Assistant La Jolla, CA
 - Developed an audio alignment pipeline to detect breathing patterns in children's speech, addressing frequency differences between children (**75-500 Hz**) and adults
 - Implemented voice quality feature extraction and fine-tuned speaker diarization on **50+** hours of children's audio, achieving **55%** improved speaker separation for multi-child stress analysis

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- [S.1] Nerem, R. R., Singh, P., Ward, C., Wang, Y. (2026). **Algorithmically Aligned Neural Agglomerative Tree Construction**. Under review at *International Conference on Learning Representations (ICLR 2027)* (submitted September 2026).
- [S.2] Singh, P.*, Mortensen, H.* et al. (2026). **The Federal NanoEHS Data Landscape: Machine-learning and Large Language Methods to Improve Data Accessibility, Interoperability and Semantic Queries**. Under review at *Academia AI and Applications* (submitted July 2026).
- [C.1] Singh, P., Chiba, A. (2024). **Can AI help predict stress in children's voices?** Presented at the UCSD Undergraduate Research Conference.
- [P.1] Singh, P.*, Mortensen, H.* et al. (2025). **NIH ICCVAM: The Federal NanoEHS Data Landscape**. Presented at the Interagency Coordinating Committee on the Validation of Alternative Methods Public Forum.

HONORS AND AWARDS

- **Ledell Family Research Scholarship** 2026
University of California San Diego
 - Second consecutive award, supporting neural algorithmic reasoning research on learnable linkage functions for hierarchical clustering
- **Ledell Family Research Scholarship** 2025
University of California San Diego
 - Awarded to 22 students from 1000+ applicants in support of undergraduate research in science and engineering
- **Virtual Intern Award (VSFS Program)** 2025
U.S. Department of State
 - Selected as 1 of 20 interns from 1400+ for contributions to federal nanomaterial research and cross-agency graph-based data integration

PROJECTS

- **AnkyloRL: Curriculum Reinforcement Learning for Behavioral Paleontology** 2026
Tools: MuJoCo, Gymnasium, Stable-Baselines3, PPO, Python [GitHub]
 - Built a **28-DOF** Ankylosaurus model with physics constraints and constants sourced from paleontology literature
 - Trained PPO through a 3-stage curriculum (balance, forage, evade) against a T-rex adversary policy from MesozoicLabs
 - Designed a **9-term** paleobiologically justified reward function, validated behavior over 100 eval episodes
- **OpenRECAP: Advantage-Conditioned Offline RL for Vision-Language-Action Policies** 2026
Tools: PyTorch, MuJoCo, LeRobot, SmolVLA, Gymnasium, Python [GitHub]
 - Adapted RECAP to fine-tune SmolVLA (**450M** params) for pick-and-place via advantage-weighted offline RL
 - Built the MuJoCo SO-101 environment and collected a **200-episode** teleoperation dataset for BC pretraining
 - Ran ablations on reward shaping, self-scoring, and curriculum to isolate which reward signal the value function requires
- **ResearchKG: Citation Network Generator and Literature Explorer** 2025
Tools: Next.js, D3.js, Semantic Scholar API, spaCy, GCP Cloud Run, Elasticsearch [GitHub]
 - Built a tool expanding one seed paper into a **1,000+** node citation network with author-influence metrics
 - Added cycle detection to surface circular citation dependencies, visualized interactively with D3.js

SKILLS

- **Research Methods:** Neural Algorithmic Reasoning, Experiment Design, Approximation Algorithms, Combinatorial Optimization, Size and Distribution-Generalization Analysis
- **Reinforcement Learning:** PPO, GRPO, Imitation of Algorithm Traces, Self-Improvement Search (ExIt, Gumbeldore), Curriculum Learning, Advantage-Conditioned Offline RL, MuJoCo / Gymnasium / Stable-Baselines3
- **Machine Learning:** PyTorch, PyTorch Geometric, HuggingFace, CUDA, JAX
- **Optimization & Scientific Computing:** Gurobi, Integer Linear Programming, Dynamic Programming, SciPy, NetworkX, NumPy
- **Programming Languages:** Python, C++, Java, JavaScript, SQL, SPARQL